

AMENDMENT NO. 1 JANUARY 1992
TO
IS 3522 (Part 2) : 1989 TEXTILES — ESTIMATION
OF COMMON PRESERVATIVES — PART 2
(First Revision)

(This amendment is being issued to incorporate estimation of permethrin, a new mothproofing agent for wool and its blends, developed by Defence Research and Development Establishment, Gwalior, Defence Research and Development Organization, Ministry of Defence, Government of India, after extensive research and field trials.)

(Page 1, clause 1.1) — Insert the following at the end of the clause:

f) Permethrin

(Page 5, clause 10.4.2) Insert the following new clauses after 10.4.2:

'11 ESTIMATION OF PERMETHRIN

11.1 General

11.1.1 This test method is applicable for estimation of permethrin applied in wool fabrics for its protection from insect damage. Permethrin is extracted from the wool fabric into solution. This extract, at a known dilution is then injected to the Gas Liquid Chromatographic apparatus (GLC) under specified conditions. The result of the analysis is then compared with standard known solution of permethrin, obtained under the same conditions.

11.2 Procedure

11.2.1 Weigh a sample of mothproofed wool fabric of 10 g after conditioning for 12 h at 27°C and 65 percent relative humidity. Transfer it to a Soxhlet extraction apparatus and extract for 6 h at a rate of six solvent exchange per hour with 100 ml of 2-methoxyethanol solvent in round bottom flask. Evaporate the extract to dryness and dissolve the extract in 5 ml of hexane and transfer to 10 ml volumetric flask. Rinse the round bottom flask with 5 ml hexane and transfer it to volumetric flask and make up to the mark.

11.2.2 To prepare standard solutions of known concentrations, dissolve accurately weighed quantity of permethrin (technical) in a suitable volume of isopropanol. From this stock solution prepare standard solutions of desired concentration by dilution with isopropanol in volumetric flask (10 ml).

11.2.3 Inject 2 or 3 µl of standard solutions (obtained in **11.2.2**) and then of extract sample (obtained in **11.2.1**) into the GLC operated under the conditions given in **11.2.4**. Thus obtain the chromatographs for standard solutions (known concentrations) and for the extract sample (unknown).

11.2.4 Operating Conditions

11.2.4.1 The operating conditions of the GLC with Flame Ionization Detector shall be as follows:

<i>Column</i>	—	Glass column, 2 percent OV-17 on chromosorb AW DMCS, 1 metre (length) × 3 mm (internal diameter)
<i>Temperatures</i>	—	Oven — 263°C Injector — 290°C Detector — 290°C
<i>Gas flow rates</i>	—	N ₂ — 76 ml/min H ₂ — 40 ml/min Air — 300 ml/min
<i>Attenuation</i>	— ×	8
<i>Chart speed</i>	—	5 mm/min
<i>Retention time</i>	—	250 seconds
<i>Analysis time</i>	—	18 minutes

NOTE — A typical gas chromatographic run showing retention time and shape of the peak is given in Fig. 1.

11.2.5 Measure the peak areas of the standard solutions of permethrin and draw a calibration graph by plotting peak areas against the known concentrations.

11.2.6 Measure the peak areas of the extract sample and convert into a concentration by using calibration graph (obtained in **11.2.5**). Then calculate quantity of permethrin present in the total extract sample (10 ml) (obtained in **11.2.1**).

11.2.7 The percentage of permethrin present on the wool weight is determined by using the following formula;

$$\text{Permethrin, percent (on wool weight)} = \frac{100 \times QPE}{WWF}$$

QPE = calculated quantity of permethrin present in the extract sample, and

WWF = weight of wool fabric taken for extraction.

11.2.8 Repeat the test with the remaining test samples and calculate the percentage of permethrin on the wool sample and then determine the average of all the values.

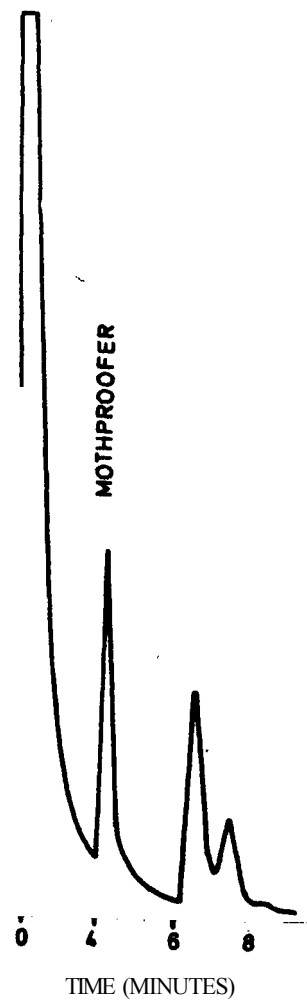


FIG. 1 GC CHROMATOGRAPH OF MOTHPROOFER

(TXD 5)